Computational Physics: Introductory Course

Assignment 5:

Differential Equations I

Per Jönsson & Jörgen Ekman
Faculty of Technology and Science
Malmö University

What the report should contain

The report should contain:

- The py-files, with carefully commented Python code, that were used to solve the exercises. All code should start with a sentence or two describing what the code is doing. Each py-file should be embedded in the LaTeX or Word document.
- When applicable, you should include the output or plots produced by your code as part of the exercise.
- When you, as part of an exercise, are asked to draw physical conclusions it is important that these conclusions are included in the report. If you are unable to draw any conclusion you should explicitly write 'I was not able to draw any conclusion'.

Important things to keep in mind for producing a good report that adheres to the general rules in scientific writing.

- All variables in the report should be in math font, e.g. the distance was $d = 1.5$ (do not write the distance was d = 1.5, wrong).
- All units in the report should be in correct font, e.g. the acceleration is $a = 1.5$ ms^{-2} (do not write the acceleration is $a = 1.5\,ms^{-2}$, wrong).
- Write $a = 1.5 \cdot 10^{-3}$ instead of $a = 1.5 * 10^{-3}$ (wrong, never use * in equations or text)
- If you give computed values, round them to a reasonable number of decimals.
- The font size in plots should be large enough for the axis texts and legends to be read.

Reading instructions for assignment 5

Read sections 21.1 - 21.5 in the lecture notes.

Exercises

1. Solve the following ODEs in $[0, 10]$ and plot the solution. Choose enough t values so that the plot is smooth, see example 18.4.

   (a) $z' = te^{-z}, \quad z(0) = -2$

   (b) $y'' + y = \sin(x), \quad y(0) = 0,\ y'(0) = 0$

   Solve (a) and (b) analytically, use Wolfram Alpha if necessary `http://www.wolframalpha.com/`. Compare the plots of the exact and the numerical solutions.

2. Solve the system of ODEs and plot the solutions in the interval $[0, 20]$. The different components should be plotted in the same figure and the command `legend` should be used to indicate which is which.

$$\begin{cases} y_0'' = -2y_0 + y_1 & y_0(0) = 10,\ y_0'(0) = 0 \\ y_1'' = y_0 - y_1, & y_1(0) = 15,\ y_1'(0) = 0 \end{cases}$$

3. A sky diver with the mass $m = 100$ kg falls with the speed $v_0 = 20$ m/s when he/she releases the parachute. The released parachute leads to a breaking force that is proportional to square of the speed of the sky diver (constant of proportion $k = 40$ Ns^2/m^2). If we choose reference direction downwards then the force on the diver is

$$F(t) = mg - kv(t)^2.$$

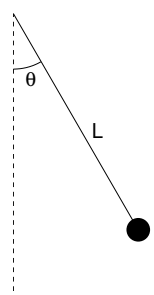
Combined with Newton's second law $F(t) = ma(t) = mv'(t)$ we get

$$\begin{cases} v' = g - \dfrac{k}{m}v^2 \\ v(0) = v_0 = 20. \end{cases}$$

Solve the problem in the time interval $[0, 10]$. Plot the solution and write suitable text on the axes Determine the speed of the sky diver when t is large.

You can determine the speed of the sky diver when t is large (terminal velocity) without solving the differential equation. Can you figure out how to do that? Hint: when the terminal velocity is reached the acceleration v' is zero.

4. Consider the pendulum



The angle θ depends on time t and satisfies the differential equation

$$\theta''(t) = -\frac{g}{L}\sin(\theta(t)),$$

where L is the length of the pendulum and g acceleration of gravity. For a derivation of the equation see physics background in assignment 2 and also
`https://en.wikipedia.org/wiki/Pendulum_(mathematics)`

(a) Let $L = 1$ and $g = 9.81$. Solve the ODE with some different initial conditions $\theta(0)$ from 0.1 to 1 (radians) and $\theta'(0) = 0$ and plot the solutions in the interval $[0, 10]$.

(b) Use the approximation $\sin(\theta) \approx \theta$ and solve the differential equation analytically. Compare the numerical solutions in (a) with the analytical solutions by plotting the two solutions together. See roughly for which initial condition $\theta(0)$ the numerical and analytical solutions start to differ, e.g. when the approximation $\sin(\theta) \approx \theta$ breaks down. It can be informative to plot $\sin(x)$ and x together and compare when the curves start to deviate.

5. Go to example 21.6 in the lecture notes and look at the trajectory of a ball.

(a) Change the eventfunction so that you solve the equation from $t = 0$ to the time when the

y-coordinate becomes 0, i.e. when the ball hits the ground. Plot your solution.

(b) When friction is neglected it is known that the longest throws will be the ones where the throwing angle is $\theta = 45^o$. Which angle θ will give the longest throw when there is friction? Does the angle depend on the throwing speed?

Assume a friction coefficient $c = 0.25$ and initial conditions $x(0) = y(0) = 0$ and $x'(t) = v\cos(\theta)$, $y'(t) = v\sin(\theta)$. Set $v = 10$ and determine the angle θ that gives the longest throw. Plot the trajectory for this angle. The angle needs to be determined only as a integer, e.g. $\theta = 34^o$ or $\theta = 52^o$ Hint: you can put the solution of the differential equation in an external loop where the angle θ goes from 1 degree to 89 degrees in steps of 1 (remember that you have to convert from degrees to radians when calling `cos` and `sin`). You need to have an if-statement to keep track of the angle that gives the longest throw.

(c) Does the angle depend on the throwing speed v? Yes, or no. Could you come up with some physical motivation for your answer?

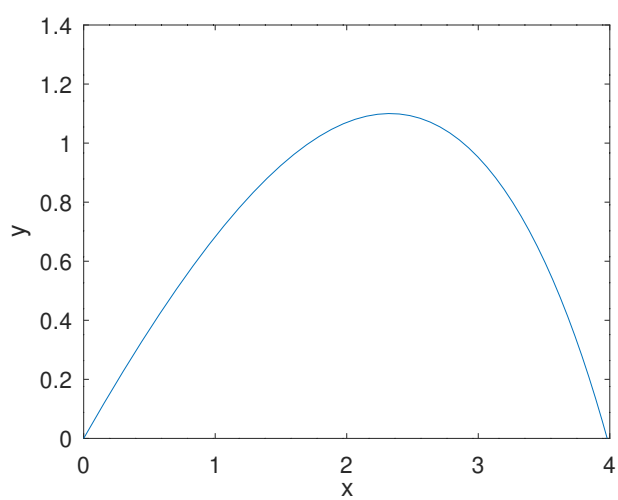
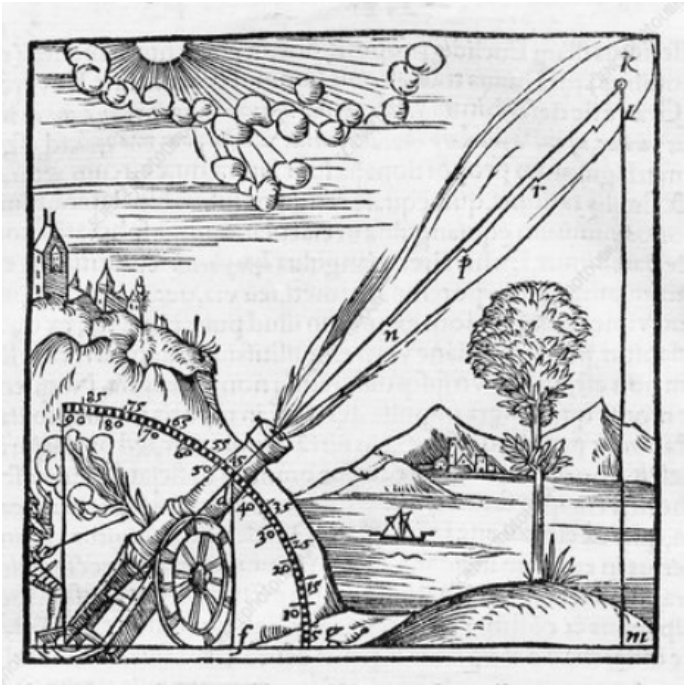
Physics background

For accurate prediction of trajectories we need to account for air resistance. Air resistance increases with the velocity of the object, and the precise form for the force $\boldsymbol{F}_{\text{air}}$ depends on the flow around the object. Commonly, it is approximated as

$$\boldsymbol{F}_{\text{air}} = -\frac{1}{2}C_d \cdot \rho \cdot A|\boldsymbol{v}|\boldsymbol{v},$$

where C_d is the drag coefficient, ρ is the density of the air, and A the cross section of the object. The drag coefficient is a dimensionless parameter that depends on the geometry of the object – the more streamlined the object, the smaller the coefficient. At speeds above approx 20 cm/s, the wake below the object develops vortex rolls, and the drag coefficient is approximately constant ($C_d \approx 0.5$ for a smooth sphere).

Without air resistance the trajectory of a projectile has a parabolic shape. Taking the air resistance into account, the trajectory is squeezed together at the second half as shown in figure below.



The squeezed trajectory historically made people think that a canon ball goes along a straight line until it's 'energy' is exhausted. At this point the canon ball is just dropping down, see the old woodcut below.